ECO 310: Precept 3

Compensated Demand

1. Assume $u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$. Find the compensated demand and the expenditure functions.

   Answer: the compensated demand bundle solves two equations:

   - allocate $ optimally (equal bang-per-buck, assuming an interior solution):

   $$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow \frac{1}{2\sqrt{x_1}} / \frac{1}{2\sqrt{x_2}} = \frac{p_1}{p_2} \Rightarrow \sqrt{x_2} = \frac{p_1}{p_2}\sqrt{x_1}$$

   - get utility $u^* : \sqrt{x_1} + \sqrt{x_2} = u^*$
   - plug the first expression into the second, and it becomes

   $$\sqrt{x_1} + \frac{p_1}{p_2}\sqrt{x_1} = u^* \Rightarrow \sqrt{x_1} = \frac{p_2}{p_1 + p_2} u^*$$

     Plugging this back into the equal bang-per-buck condition, $\sqrt{x_2} = \frac{p_1}{p_1+p_2} u^*$. Squaring both of these terms gives the actual compensated bundle:

   $$(x_1^c, x_2^c) = \left(\left(\frac{p_2}{p_1 + p_2} u^* \right)^2, \left(\frac{p_1}{p_1 + p_2} u^* \right)^2 \right) \tag{1}$$

   - and the expenditure function is the cost of the compensated bundle:

   $$\begin{aligned} e(p_1, p_2, u^*) &= p_1 x_1^c + p_2 x_2^c \\ &= p_1 \left(\frac{p_2}{p_1 + p_2} \right)^2 (u^*)^2 + p_2 \left(\frac{p_1}{p_1 + p_2} \right)^2 (u^*)^2 \\ &= \frac{p_1 p_2^2 + p_2 p_1^2}{(p_1 + p_2)^2} (u^*)^2 \\ &= \frac{p_1 p_2}{p_1 + p_2} (u^*)^2 \end{aligned} \tag{2}$$

Income and Substitution Effects

For the example in Q1, with $u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$, find the substitution effect, compensating variation, and income effects if $p_2 = 1$, and p_1 increases from 1 to 2 while $m = 6$. Explain what this means in words. You may take as given that the "regular" (Marshallian) demand function is as follows (this is what you'd get by solving (1) $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ and (2) $p_1 x_1 + p_2 x_2 = m$):

$$(x_1, x_2)(p_1, p_2, m) = \left(\frac{p_2}{p_1} \frac{m}{p_1 + p_2}, \frac{p_1}{p_2} \frac{m}{p_1 + p_2} \right) \tag{3}$$

Answer:

- first, let's find the original bundle (at old prices and income, $p_1 = p_2 = 1$ and $m = 6$): plugging into (3), it is
$$(x_1^o, x_2^o) = (3, 3)$$

- now, let's find original utility: plugging (3,3) into the utility function, it is $u^o = \sqrt{3} + \sqrt{3} = 2\sqrt{3}$
- next, let's find the final bundle (at new prices and old income, $p_1' = 2, p_2 = 1, m = 6$): plugging into (3), it is
$$(x_1^f, x_2^f) = \left(\frac{1}{2}\frac{6}{2+1}, \frac{2}{1}\frac{6}{2+1}\right) = (1, 4)$$
- next, let's find the compensated bundle: what most efficiently gets old utility $2\sqrt{3}$ at new prices. Plugging $p' = 2, p_2 = 1, u^* = u^0 = 2\sqrt{3}$ into (1), it is
$$\left[\left(\frac{p_2}{p_1' + p_2}u^*\right)^2, \left(\frac{p_1}{p_1' + p_2}u^*\right)^2\right]_{p_1'=2, p_2=1, u^*=2\sqrt{3}} = \left[\left(\frac{1}{2+1}\cdot 2\sqrt{3}\right)^2, \left(\frac{2}{2+1}2\sqrt{3}\right)^2\right] = \left(\frac{4}{3}, \frac{16}{3}\right)$$
- now, the substitution effect is $x_1^c - x_1^0 = \frac{4}{3} - 3 = -1.6667$, and the income effect is $x_1^f - x_1^c = 1 - \frac{4}{3} = -0.33333$
- interpretation: ultimately, the price increase drops the consumer's good 1 purchase from $x_1^0 = 3$ to $x_1^f = 2$: 1.667 units of this is due to reoptimization (even if subsidized alongside the price increase to maintain original utility, there would be a smarter way to do it); and the rest is due to income effect: in fact nobody is offering a subsidy, so he cannot afford to maintain old utility
- finally, the CV is the subsidy needed to maintain old utility at new prices: plugging $p_1' = 2, p_2 = 1, u^* = 2\sqrt{3}$ into (2), the income now needed to get old utility is
$$\left[\frac{p_1 p_2}{p_1 + p_2}(u^*)^2\right]_{p_1=2, p_2=1, u^*=2\sqrt{3}} = \frac{2\cdot 1}{2+1}(2\sqrt{3})^2 = \frac{2}{3}(4\cdot 3) = \$8$$

  But the consumer's actual income is \$6. So he needs a \$2 subsidy.

If Time Left Over: Expected Utility

Sally has preferences over lotteries on $\{0, \$1, \$5\}$. Show that if the independence axiom holds, and if she feels $(0,1,0) \succ (.2, 0, .8)$, then she also feels $(0,1,0) \succ (.1, .5, .4)$. Hint: consider mixing each lottery with $(0,1,0)$.

Answer: the independence axiom says that if $p \succ q$, then for any number α and lottery r, $\alpha p + (1-\alpha)r \succ \alpha q + (1-\alpha)r$. So here, let $p = (0,1,0)$,$q = (.2, 0, .8)$, $r = (0,1,0)$. By independence,

$$\begin{aligned} \alpha p + (1-\alpha)r &\succ \alpha q + (1-\alpha)r \\ \alpha(0,1,0) + (1-\alpha)(0,1,0) &\succ \alpha(.2,0,.8) + (1-\alpha)(0,1,0) \\ (0,1,0) &> (.2\alpha, 1-\alpha, .8\alpha) \end{aligned}$$

This gives the desired conclusion with $\alpha = \frac{1}{2}$ – so that $.2\alpha = .1$, $1 - \alpha = .5$, and $8\alpha = .4$.